

Perspective

Monolayer or not: Assembly configuration of SAMs in perovskite photovoltaics

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CONTEXT & SCALE The global transition toward sustainable energy has intensified the search for next-generation photovoltaic technologies. Metal halide perovskite solar cells (PSCs) have emerged as a promising photovoltaic material, combining outstanding optoelectronic performance with low-cost, solution-based manufacturing. Inverted (p-i-n) PSCs have attracted particular attention for their superior attributes, such as operational stability and compatibility with tandem integration. The advancement of high-performance inverted PSCs has been propelled by the development of a specific class of hole-selective layers, originally known as self-assembled monolayers (SAMs). These SAMs have become critical components in state-of-the-art high-performance devices. Their flexible, molecular-level tunability offers a powerful means to control interfacial energetics and suppress nonradiative recombination, which are critical to the continued efficiency gains in PSCs. The functionality of SAMs is greatly influenced by both the chemical structures and stacking configurations of the molecules. A comprehensive understanding of this microstructure is therefore essential not only for advancing perovskite technologies but also for establishing general molecular design principles for semiconductor interfaces in the broader field of molecular electronics. In this perspective, we summarize recent progress in SAM-based inverted PSCs, discuss their possible microstructural configurations under realistic processing conditions, and outline future research directions to better resolve and control these molecular interlayers toward more efficient and stable solar energy conversion.

SUMMARY

Perovskite solar cells (PSCs) have emerged as a promising next-generation photovoltaic technology due to their ease of fabrication, low cost, and exceptional optoelectronic properties. Inverted (p-i-n) PSCs have attracted particular attention for their superior operational stability and compatibility with tandem devices. Their rapid progress has been driven by the development of a specific class of hole-selective layers, originally known as self-assembled monolayers (SAMs). These organic molecules self-assemble into a molecular layer at the interface between transparent electrodes and perovskite absorbers, facilitating efficient charge extraction, and have become an essential component of high-performance devices. The efficacy of such SAM-based contacts is intimately linked to their molecular structure and packing configuration, which govern both electronic properties and the long-term stability of the interface. Despite their widespread use in high-performance PSCs, the precise molecular-level microstructures of SAM-based interlayers remain under active debate, particularly regarding whether the SAMs can indeed form a monolayer under practical processing conditions. In this context, our perspective starts with an overview of recent progress in SAM-based contacts for inverted PSCs, followed by a detailed analysis of their potential molecular assembly configurations under practical fabrication conditions. Finally, we discuss whether forming a perfect monolayer is necessary to achieve high performance and stability in thin-film photovoltaic devices.

INTRODUCTION

Halide perovskite materials have attracted significant attention over the past decade because of their exceptional optoelectronic properties.^{1,2} Perovskite solar cells (PSCs) are typically constructed in a multilayer architecture, where the light-absorbing perovskite layer is sandwiched between charge-transport layers, specifically an electron-transport layer and a hole-transport layer. According to the direction of current flow, two primary device configurations have been established: the regular (n-i-p) and the inverted (p-i-n) structure. Among them, inverted PSCs have garnered particular interest due to their reduced *J-V* hysteresis, superior operational stability, and facile tandem integration capability, making them promising for commercialization.^{3–5} The rapid progress in inverted PSCs has been largely facilitated by advances in a class of ultrathin hole-selective molecular layers, originally known as self-assembled monolayers (SAMs).^{6–8}

A typical SAM molecule consists of an anchoring group, a linker group, and a terminal group. These molecules can self-assemble on the surface of metal oxide substrates, forming an interlayer between the electrode and perovskites that facilitates selective hole transport.^{9,10} Beyond charge selectivity, well-designed SAMs can also contribute to defect passivation of the perovskite layer on top, thereby suppressing non-radiative recombination and enhancing device performance.^{11,12} Compared with conventional polymeric and metal oxide hole-transport materials, SAMs offer multiple advantages, including facile fabrication, low material consumption, minimal parasitic absorption, low resistance, high structural tunability, and the potential to achieve conformal and uniform coatings for rough substrates.^{13–15} These merits collectively make SAM-based contacts a critical component in state-of-the-art inverted PSCs. Their performance is intimately linked not only to the chemical structure of the constituent molecules but also to how molecules assemble and stack on the substrate surface. From a molecular-design perspective, subtle variations in anchoring groups, backbone conjugation, and terminal group structures tune key properties such as binding strength, molecular dipole moment, and intrinsic molecular stability. These characteristics, in turn, dictate how molecules pack, orient, and interact at the oxide interface, ultimately determining the physical and chemical integrity of the SAM layer. It is therefore crucial to elucidate how molecular chemical structure and assembly configurations influence the stability and functionality of the SAM layer so as to provide design principles for the development of more efficient and stable SAM-based contacts for perovskite photovoltaics.

Although SAMs are widely used in high-efficiency inverted PSCs and are traditionally discussed within a monolayer framework, their actual assembly microstructure under practical processing conditions has yet to be well established. The “monolayer” terminology in PSCs largely traces back to classical surface-science SAM systems (e.g., Au-thiol), where prolonged adsorption and equilibration on atomically smooth substrates yield close-packed, single-molecule-thick assemblies. In contrast, SAMs in PSCs are typically deposited on hydroxyl-rich metal oxides under solution-processing conditions, where rapid solvent removal and substrate heterogeneity can lead to

more complex assembly configurations.¹⁶ An emerging question is whether SAMs indeed form a uniform, defect-free monolayer in practice or instead adopt multilayered stacking or a nominal monolayer with localized disordered aggregates that deviate from the ideal monolayer picture. This underscores the need to revisit the fundamental structure-property relationships in SAM-based perovskite devices. Herein, we begin by summarizing recent advances in SAM-based inverted PSCs, highlighting how molecular engineering has driven rapid efficiency progress. We then critically examine possible molecular stacking modes under practical fabrication conditions and reassess the long-standing monolayer assumption. Next, we discuss whether a perfectly ordered monolayer is truly essential to achieve high performance and operational stability in perovskite photovoltaics. Finally, we outline future research directions toward the rational design and precise control of molecular interlayers.

RECENT PROGRESS OF SAMs IN INVERTED PSCs

For years, inverted p-i-n-structured perovskite solar cells have primarily relied on hole-transporting materials such as (poly(3,4-ethylenedioxythiophene):polystyrenesulfonate) (PEDOT:PSS), (poly[bis(4-phenyl)(2,4,6-trimethylphenyl)amine]) (PTAA), and NiO_x. Despite their early success, these materials are plagued by several practical issues, including chemical instability, corrosive characteristics, unfavorable energy alignment, poor wettability, and undesired reactivity with the perovskite layer.^{17–19} In this context, SAMs have emerged as a promising alternative to overcome these limitations.

SAMs feature a molecular architecture typically composed of an anchoring group, a spacer or linker group, and a terminal group. The anchoring groups interact with hydroxyl-rich electrode surfaces, allowing the molecules to assemble into a molecular layer at the interface between transparent electrodes and perovskite films.²⁰ The spacer groups, which connect the anchoring and terminal groups, contribute to regulating the molecular packing, solubility, and conformational freedom of the SAMs.²¹ The terminal groups are often regarded as the most influential components, as they primarily govern the interfacial properties of the SAM layer. This, in turn, modulates the perovskite film quality, energy-level alignment, hole-transport efficiency, and interfacial recombination landscape.^{14,22}

Recent progress in SAMs for inverted PSCs has been primarily driven by the modification of terminal groups (Figure 1A). Building on the early demonstration of SAM-type hole-transporting materials, such as V1036,¹⁴ used in inverted PSCs, carbazole-based derivatives soon prevailed in the field with enhanced PCEs.²³ Further modification of carbazole-based core groups centers around several aspects, including heteroatom insertion,^{24–26} side-chain engineering,^{23,27–29} and conjugation extension.^{7,30,31} These strategies can influence intermolecular interactions during assembly and potentially modulate molecular packing, enabling multifaceted optimization of the SAM layer, including interfacial energetics and the carrier recombination landscapes. Besides carbazole, triphenylamine (TPA), with a non-planar conformation and strong electron-donating character, serves as another widely adopted core in SAM design.

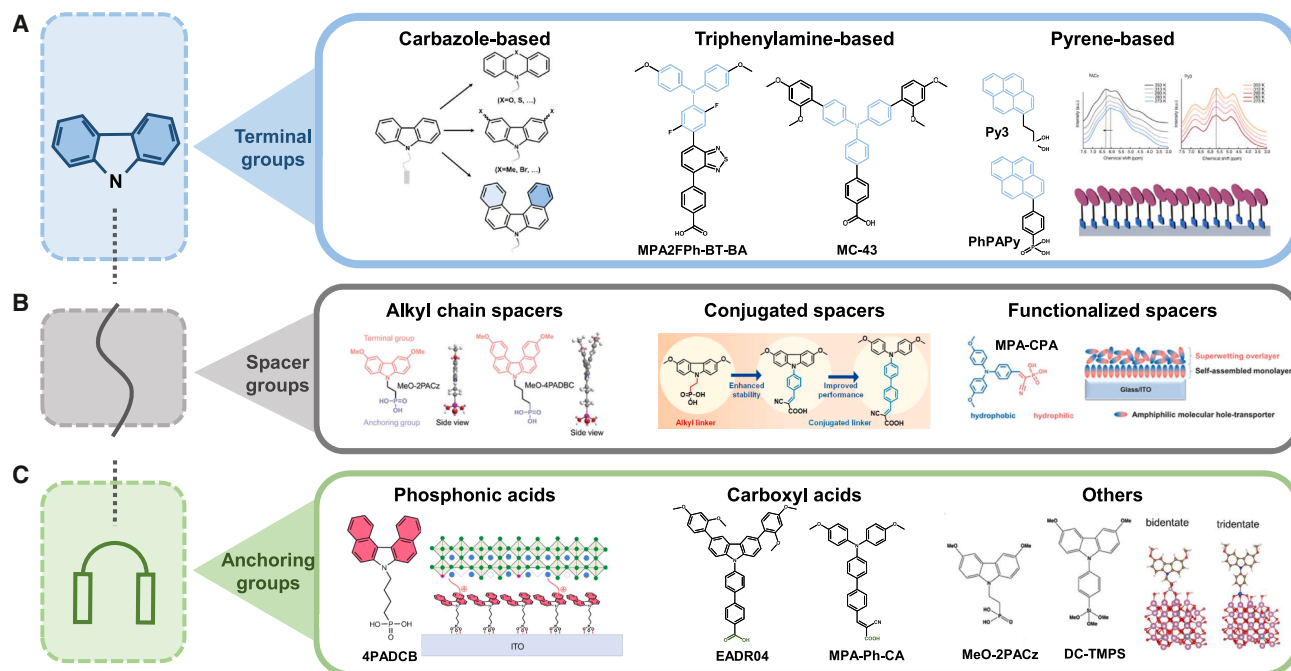


Figure 1. Structural components of SAMs and an overview of molecular design strategies

(A) Terminal groups: representative conjugated cores, including carbazole-, triphenylamine-, phenothiazine-based moieties, as well as other aromatic or heteroaromatic cores.^{9,35–37}
 (B) Spacer (linker) groups: structural segments connecting the anchoring and terminal moieties, where alkyl or conjugated linkers regulate molecular packing, orientation, and charge-transport pathways within the SAM layer.^{32,38,39}
 (C) Anchoring groups: functional tail groups, including phosphonic acid, carboxylic acid, and other anchoring groups, binding to oxide substrates to ensure molecular immobilization and interfacial robustness.^{9,40,41}

Based on existing reports, it seems that TPA-based SAMs generally deliver high performance primarily in substantially modified designs, whereas bare TPA cores are less commonly associated with top-performing devices.^{32,33} These observations seem to imply that TPA itself might be suboptimal as a SAM core, thereby making further molecular modification advantageous. In this context, TPA is sometimes used as a modular building block for higher-order architectures, with dual-core designs representing a successful attempt, suggesting that structural modification could compensate for the possible limitations of the bare TPA core.³⁴ To date, such dual-core design has been dominated by carbazole-TPA combinations, which reflect the relatively limited core motifs in the SAM core reservoir. Expanding the core motif library could therefore enable a broader range of multi-core designs, potentially extending beyond current dual-core motifs, and further enlarge the molecular design space for future SAM development.

In this context, a distinct class of peri-fused polyaromatic cores (e.g., pyrene-based derivatives) has been very recently reported. Different from the commonly explored heteroatom-substituted structures, these all-carbon compounds provide rigid frameworks, with high chemical inertness and structural rigidity,^{8,35,36,42} which further expand the terminal group reservoir.

Spacer group designs have mainly focused on alkyl chain spacers, conjugated spacers, and functionalized spacers, serving as a bridge connecting the anchoring and terminal units, which fa-

cilitates the regulation of properties such as molecular packing, interfacial energetics, and charge transport (Figure 1B). Alkyl chain spacers, composed of saturated carbon chains, offer excellent processability and conformational flexibility to accommodate interfacial roughness.³⁹ By varying the chain length, molecular orientation and inter-molecular interactions can be tuned, with shorter alkyl chains (C2–C4) promoting compact packing and efficient carrier transport and longer chains (C4–C6) providing greater flexibility for molecular stacking and potentially introducing additional tunneling resistance.^{23,24,38,43} Conjugated spacers, such as phenyl or thiophene linkers, extend π -conjugation toward the anchoring group, strengthening electronic coupling and facilitating more delocalized hole transport across the interface.^{37,44,45} Functionalized spacers generally incorporate electron-donating or -withdrawing motifs within the linker, which can reshape intramolecular charge distribution rather than serving only as a geometric linker. They can also improve wetting during perovskite deposition and, in some cases, introduce specific interfacial interactions that aid defect passivation.^{20,32,46}

The anchoring group plays a decisive role in dictating molecular anchoring, as well as contributing to tuning molecular orientation and long-term stability of SAMs (Figure 1C). In perovskite photovoltaics, phosphonic acid anchors have become the dominant motif, as they can form strong bi- or tridentate coordination with metal oxides and markedly improve interfacial robustness.^{47,48} Beyond phosphonates, alternative anchoring chemistries have

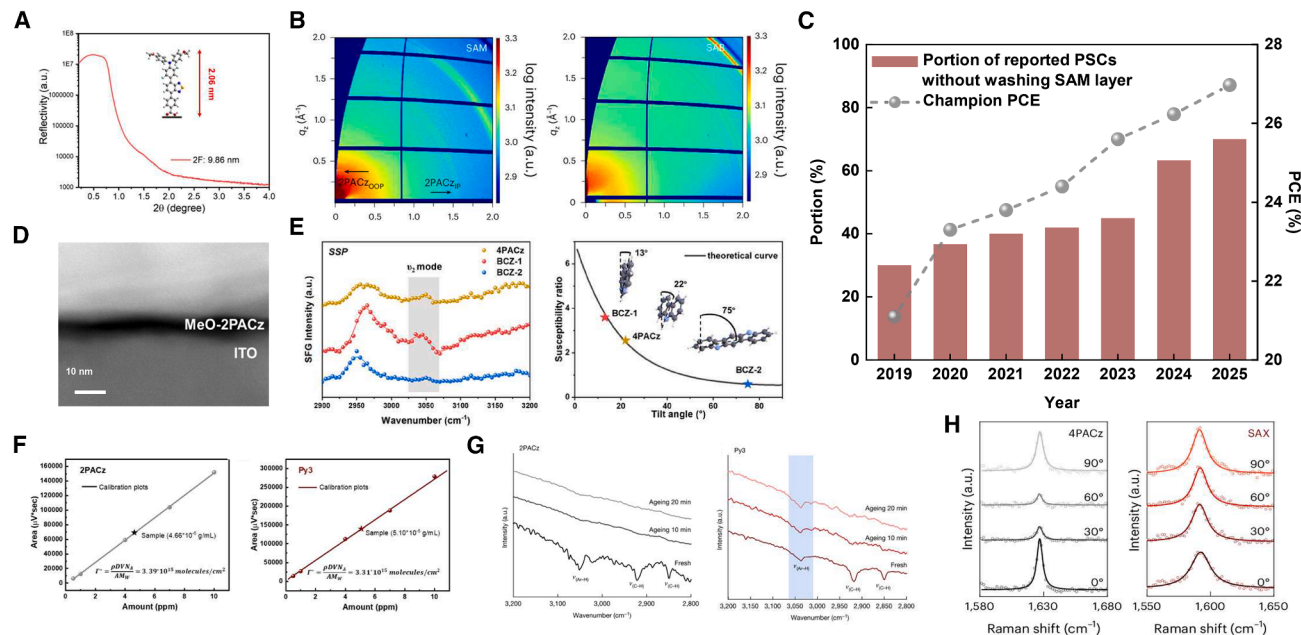


Figure 2. Characterizations of SAM layers

- (A) XRR spectrum of the 2F deposited on ITO glass, with the inset showing the theoretically calculated single-molecule height.⁹
 (B) Two-dimensional GIWAXS patterns of 2PACz and SAB. IP, in-plane; OOP, out-of-plane. Diffraction signals with $q > 2.0 \text{ \AA}^{-1}$ are attributed to ITO substrates. The schematic on the right depicts the structural configuration of a self-assembled bilayer.⁵⁵
 (C) Portion of reported inverted perovskite solar cells fabricated without solvent rinsing as post-treatment, and the corresponding champion power-conversion efficiency over time.
 (D) Cross-sectional high-angle annular dark-field (HAADF) image of the MeO-2PACz layer.⁴³
 (E) The ssp-polarized SFG spectra for quartz/SAMs, where the SAMs are 4PACz, BCZ-1, and BCZ-2. The gray box marks the C–H ν_2 stretching vibration peak of the benzene ring. Theoretical curve of the susceptibility ratio as a function of the tilt angle θ is shown alongside experimentally determined angles and schematic molecular orientations.⁵⁸
 (F) Calibration plots and the determined concentration of 2PACz and Py3 in the UPLC measurements.³⁵
 (G) Polarized FTIR spectra measured with p-polarization for the fresh and aged films of 2PACz and Py3.³⁵
 (H) Polarization-dependent Raman spectroscopy of the 4PACz and SAX films.³⁸

also been explored. Although carboxylic-acid anchors are attractive because of their facile synthesis and good solubility in polar solvents, studies suggest that their predominantly monodentate binding and relatively limited hydrolytic stability can result in less robust attachment on oxide surfaces.^{20,40,49,50} To address these potential limitations, –COOH groups are in some cases paired with a strongly electron-withdrawing –CN to form cyanoacrylic acid groups or integrated with a conjugated spacer to improve molecular rigidity and packing.^{13,51} Other anchoring groups, such as boric-acid-based anchoring groups, have been reported to attenuate interfacial acidity and reduce substrate corrosion,^{34,52,53} and silane derivatives enable tridentate covalent binding with the substrate, offering a more chemically resilient interfacial linkage.⁴³

ASSEMBLY CONFIGURATIONS OF SAMs

Although recent advances have largely centered on molecular design and how their chemical structures influence device performance, the assembly configuration of SAMs, including how molecules pack, orient, and interact at the oxide interface, is more complex and remains relatively elusive. Despite being conventionally described as a monolayer, multiple

experimental observations suggest that practical assembly structures of SAM films can deviate from an idealized monolayer assembly model. For instance, X-ray reflectivity (XRR) measurements sometimes report molecular-layer thicknesses of up to $\sim 10 \text{ nm}$, substantially exceeding that of a single molecular layer^{9,54} (Figure 2A). Electrochemical estimates based on cyclic voltammetry (CV) have reported surface molecular densities in the order of 10^{14} per cm^2 , nearly an order of magnitude higher than expected for an ideal monolayer.⁵⁰ Grazing-incidence wide-angle X-ray scattering (GIWAXS) patterns of SAM layers exhibit distinct diffraction features, revealing that these films are, at least partially, crystallized with long-range out-of-plane periodicity and thus adopt a multiple-layer configuration rather than a single molecular monolayer^{43,55–57} (Figure 2B). These results indicate that, under realistic processing conditions, SAM layers may comprise partially disordered aggregated domains (often acting as defects) or multilayered structures (ordered structures that may support charge transport) rather than a strictly defined monolayer. This raises a fundamental question about the actual molecular assembly: are SAMs in practice truly uniform monolayers with full substrate coverage and aligned orientation, as idealized in Figure 3A, or are they more accurately described

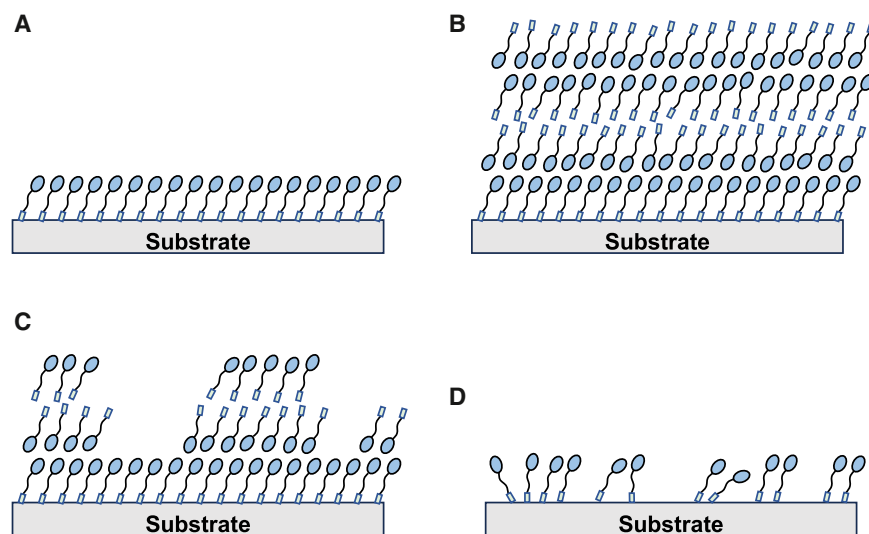


Figure 3. Schematic illustration of possible molecular assembly configurations in SAM-based films on oxide substrates under different processing conditions

(A) Idealized monolayer assembly where molecules uniformly anchor on the substrate with close-packed ordering.

(B) Unrinsed film featuring a large number of weakly adsorbed molecules, giving rise to a multilayer assembly.

(C) SAM film after typical post-treatment practices, where loosely bound species are partially removed, resulting in a thinner yet heterogeneous assembly with residual multilayers on top of the anchored layer.

(D) SAM film after aggressive post-treatment, potentially leading to disturbed packing and partial surface exposure.

as complex assemblies comprising multiple molecular layers or aggregated domains?

The practical assembly structure of SAMs is strongly influenced by their fabrication and processing conditions, such as deposition method and post-treatment procedures. SAMs in PSCs are typically processed through solution-based methods, such as dip coating or spin coating.¹³ For the as-fabricated films, a post-treatment step, such as solvent rinsing, is sometimes applied to remove physisorbed molecules and promote molecular ordering. Early studies commonly adopted such rinsing steps to eliminate unbound molecules,^{59–61} reflecting the prevailing assumption that a monolayer configuration was favored for achieving high device performance. More recently, however, an increasing number of high-performance SAM-based devices have been reported without solvent rinsing^{33,61–63} (Figure 2C), indicating that high performance can also be achieved without post-treatment and prompting closer examination of the assembly structures produced under different processing conditions.

As evidenced by the characterization results discussed above, SAM layers fabricated without any post-treatment adopt a multiple-layer configuration rather than a single molecular monolayer. Such stacking configurations are further supported by high-angle annular dark-field (HAADF) imaging, which directly visualizes an interfacial layer with a thickness of 8–10 nm, far exceeding the molecular length expected for a single layer^{49,64} (Figure 2D). In addition, contact-angle measurements before and after solvent rinsing show pronounced changes in surface wettability, indicating that the rinsing process removes a portion of loosely bound molecular species from the surface.^{49,65} Notably, sum-frequency generation (SFG) spectroscopy, a nonlinear optical technique, selectively probes the non-centrosymmetric interface, providing direct evidence of the formation of an anchored and ordered monolayer at the buried oxide substrate interface^{35,42,66} (Figure 2E). Therefore, these complementary characterization results collectively point toward a practical assembly structure in which the SAM films comprise an ordered monolayer at the buried interface, with additional molecular

layers stacked on top, forming a multi-molecular assembly characteristic of non-rinsed samples (Figure 3B).

For SAM layers subjected to solvent rinsing as a post-treatment, while not denying that sophisticatedly controlled film deposition and post-treatment conditions could yield a monolayer or approach monolayer thickness, as indicated by the XRR-measured thickness of 0.6–2 nm in a few studies,^{53,67} they generally exhibit characteristics consistent with multilayered assemblies. For instance, ultra-performance liquid chromatography (UPLC) is used to quantify the density of the molecules in the as-rinsed molecular films.³⁵ The molecules were removed from the ITO substrates and dissolved in methanol for the UPLC measurements. The densities of the molecules were determined to be about an order of magnitude higher than those expected for a monolayer configuration (Figure 2F). Meanwhile, it has been reported that the conventionally adopted washing method, using ~200 μ L of solvent to dynamically rinse the film, is not enough to remove all the weakly adsorbed molecules, as shown in the evolution of molecular coverage by XPS.⁴³ Ellipsometry measurements on rinsed films also frequently yield thicknesses larger than those expected for a single molecular layer.^{31,38,68} More importantly, for the washed SAM films, GIWAXS measurements sometimes also showed distinct diffraction features along both q_{xy} and q_z directions, suggesting both in-plane and out-of-plane long-range periodicity of molecular stacking.^{41,55–57} The presence of such long-range ordering also indicates that the SAM layer cannot be a single-molecular monolayer but rather involves multilayer or partial molecular aggregations that are crystallized.⁵⁸ Although solvent rinsing can reduce weakly physisorbed molecules, and may yield near-monolayer coverage in some regions, the observed crystalline nature, surface morphology, and measured thicknesses suggest that the post-treated films may still retain multilayered components. Accordingly, describing these films as true monolayers may not be strictly accurate (Figure 3C). Although in principle one could attempt to remove all the weakly adsorbed molecules by applying more aggressive post-treatment, such as rinsing with large solvent volumes or performing multiple rinsing cycles,

such harsh procedures may expose the underlying ITO surface and disrupt molecular packing. This, in turn, would compromise the interfacial quality and degrade device performance (Figure 3D). Under such conditions, the formation of a monolayer configuration would be achieved at the expense of interfacial integrity.

Further insights into the assembly microstructures of SAMs can be obtained from molecular orientation analyses by polarization-dependent spectroscopic techniques. Specifically, SFG with different polarization combinations offers surface-specific sensitivity to the orientation of the molecules at the buried interface.⁵⁹ The results generally suggest that the molecules may not be perfectly upright but adopt a moderately tilted configuration relative to the substrate, implying that the effective vertical dimension of the molecular layer could be slightly smaller than the fully extended molecular length (~1 nm). Accordingly, the reported film thicknesses of ~1–2 nm for rinsed SAMs, although comparable to those of a nominal monolayer, may not necessarily correspond to ideal monolayer formation. These measured thicknesses could be affected by factors such as molecular tilt, localized monolayer stacking, or experimental uncertainties in ellipsometry and XRR measurements of ultrathin organic films.⁶⁹ Even when certain regions appear to have a monolayer-like configuration under high-resolution characterizations, achieving complete uniformity across the entire surface remains challenging. The possible coexistence of lateral microstructural domains, such as locally aggregated regions or multiple layers, suggests that the SAM film may deviate from a strictly defined, perfectly uniform monolayer.

In addition to SFG, complementary techniques, such as GIWAXS, polarized Fourier transform infrared (FTIR), and Raman spectroscopies, have also been employed to study the molecular orientation in SAMs^{35,38,58} (Figures 2G and 2H). For a true monolayer, however, these techniques are not likely to be applicable, as the inherently weak vibrational or scattering signals from a single molecular layer render its reliable detection challenging. In the case of multilayer or partially aggregated structures, these techniques instead yield ensemble-averaged information integrated over the entire film thickness, offering limited capability to resolve how molecular orientation evolves vertically or varies laterally across domains. Whether the molecular tilt angle or stacking geometry changes gradually from the substrate interface to the uppermost surface remains unclear due to the lack of depth-resolved orientation sensitivity. Likewise, spatial heterogeneity in molecular packing and orientation across the surface has scarcely been mapped. Although such lateral variations may influence perovskite nucleation and buried-interface uniformity by creating spatially nonuniform wetting and electrostatic landscapes, current techniques often lack the spatial and depth resolution required for systematic mapping. These challenges underscore the need for multi-modal, depth-sensitive characterization approaches to construct a more complete understanding of SAM assembly configurations.

Given the complexity of SAM assembly configurations discussed above, it may be worth considering a more universal term, such as self-assembled molecular layer, instead of SAM, to better reflect the diversity of molecular assembly configurations in the films subjected to realistic processing conditions.

IS A MONOLAYER A NECESSITY?

Both rinsed and unrinsed SAM layers have been reported to contribute to high device efficiencies,^{35,60,69} with unrinsed SAMs becoming increasingly common in recent years. These observations have raised the question of whether the formation of a perfectly defined monolayer is truly necessary, or even ideal, for achieving optimal device performance.

From an efficiency perspective, forming a well-defined monolayer may help minimize parasitic absorption due to its monolayer thickness. Meanwhile, the uniform molecular coverage may reduce interfacial defects and can promote homogeneous perovskite crystallization. In addition, their ordered molecular arrangement may allow fine-tuning of surface dipoles and offer a versatile platform for targeted structural functionalization, potentially enabling defect passivation at the buried perovskite interface.^{70–72} For the multilayered assemblies formed without post-treatment, it has been reported that they improve substrate wettability for subsequent perovskite deposition, as a greater number of hydrophilic anchoring groups can be exposed at the surface.^{26,73} Such multilayered configurations, although generally less smooth than monolayer structures, do not necessarily compromise perovskite film quality because defect formation is also influenced by the underlying surface chemistry, such as wettability.⁷⁴ In addition, the free phosphonic acid groups within the stacked molecular layers have been reported to interact with FAI, which is considered to be beneficial for stabilizing the black phase of the perovskite. Given that these assemblies are only a few nanometers thick, parasitic absorption should not pose a significant concern.^{46,74,75} Taken together, both monolayer and multilayer structures can yield high-efficiency devices. Considering devices based on untreated SAM films already yield high efficiency, the formation of a perfectly defined monolayer does not appear to be a prerequisite for realizing high efficiency. Nevertheless, these processing and initial efficiency benefits may entail a stability trade-off due to the weakly adsorbed molecules.

From a stability perspective, if every single molecule can be securely anchored to the substrate, a well-ordered monolayer should, in principle, offer superior film stability compared with its multilayer counterpart. This is because the excess unanchored or weakly bound organic molecules in the multilayered structure are among the most mobile species in the device, making them highly susceptible to structural rearrangement or diffusion under external stress, such as an electric field. Such structural evolution can progressively reshape the interfacial stacking configuration, leading to drift in local interfacial energetics. Meanwhile, the disordered aggregates may increase contact resistance and create additional interfacial trap sites, leading to accelerated operational degradation. Environmental stimuli further amplify these processes. For example, moisture exposure can weaken molecular anchoring at the SAM-conductive oxide interface through the hydrolytic cleavage of the P–O bond.⁷⁶ This can facilitate molecular desorption or rearrangement and resultantly drive the molecular stacking configuration to deviate from the ideal monolayer structure. Light exposure, particularly in the ultraviolet range, could introduce relaxation of SAM stacking configurations by introducing energetic

perturbations comparable to typical SAM-substrate binding energies (~ 3 eV), thereby rendering interfacial bonding and molecular arrangements susceptible to disruption or rearrangement.^{77,78} Such molecular reorientation, stacking collapse, or migration can therefore compromise the structural integrity of the SAM layer and deteriorate device performance. Although strengthened intermolecular interactions, such as π - π stacking, can be achieved through careful molecular design to improve packing robustness, these noncovalent forces remain much weaker than the covalent anchoring between anchoring groups and the oxide substrate. Therefore, a perfect monolayer, with every molecule chemically bound to the surface, is expected to provide inherently greater structural and thermal stability. In the future, however, if robust covalent anchoring can be extended to all molecules within a multilayer structure, such multilayered configurations might achieve comparable stability.

From an electronic functional perspective, a mono-molecular layer, as reported in many studies, cannot be strictly regarded as a conventional charge-transport layer. It is more likely to act as an energy-level modifier.⁷⁶ A well-defined monolayer can approximate a densely packed dipole sheet, producing a relatively uniform work-function shift and more predictable interfacial energetics, whereas multilayer assemblies may partially average or locally compensate dipoles due to reduced orientational order. In the case of a genuinely ultrathin monolayer, carrier transport is therefore often discussed within the tunneling framework.^{79–81} As the assembly transitions to multilayered structures of ~ 10 nm, it is sometimes assumed that non-resonant tunneling alone would be insufficient over the full thickness and that charge transport would require intermolecularly coupled routes, with thermally activated hopping considered as one plausible contributor. In this regard, packing features with enhanced short-range electronic coupling (e.g., favorable π - π interactions along the transport direction) and a reasonably continuous vertical transport network could be beneficial for reducing additional series resistance. In this context, the more efficient configuration depends on whether tunneling across an ultrathin monolayer is more efficient than transport through a few-nanometers-thick, π - π -coupled overlayer. If achieving intrinsically high mobility in the molecular stack proves challenging, the monolayer configuration might remain the more favorable choice. Several studies have shown that omitting the post-deposition rinsing step does not degrade device performance,^{23,82} suggesting that the presence of weakly physisorbed overlayers does not necessarily hinder charge transport or disrupt interfacial energetic alignment in these SAM-based systems.^{24,83,84} The efficiency of carrier extraction is critically dependent on the interlayer's combined role in promoting extraction and suppressing recombination. Given that carrier transport in such multilayer molecular assemblies is not yet fully resolved, future studies clarifying that operative transport mechanisms would provide a clearer picture for molecular optimization and may substantially revise the structure-transport correlations proposed above.

From a manufacturing perspective, although a perfectly ordered monolayer can, in theory, deliver optimal interface control, its realization requires sophisticated process conditions that are difficult to reproduce in large-scale fabrication.

techniques, such as blade coating or slot-die coating, which involve a rather quick fabrication duration, do not allow enough time for the molecules to rearrange and therefore achieve a perfectly uniform monolayer. Therefore, monolayer formation across large areas may not be either feasible or necessary. These processes inherently involve dynamic solvent flow, rapid drying, and substrate roughness that hinder the formation of a strictly single-molecule layer. On rough substrates (e.g., FTO or textured silicon), the peak-valley topography and high-curvature sidewalls further make conformal monolayer coverage challenging. A nominally “perfect” monolayer may leave local uncovered pinholes that act as leakage pathways, compromising yield and reproducibility. In such cases, a few-nanometers-thick multilayer assembly may be practically preferable, as the added thickness increases tolerance to topographic variations and helps ensure pinhole-free coverage. Also, from an industrialized manufacturing standpoint, the spontaneous formation of a few-nanometers-thick SAM layer without requiring delicate post-treatment steps reduces process complexity, improves reproducibility, and enhances compatibility with roll-to-roll or continuous coating lines.^{28,85}

Overall, while monolayers remain a useful model for studying interfacial energetics, multi-molecular assemblies may provide a more manufacturable, scalable, and robust design paradigm for industrial perovskite photovoltaics. Further work that correlates assembly structure with operational stability and charge-transport pathways would be helpful to clarify and define when each configuration is more appropriate.

OUTLOOK AND PERSPECTIVES

Recent studies on SAMs have refined our understanding of SAM assembly structures in perovskite photovoltaics, revealing that their molecular stacking is far more complex than a simple monolayer and requires case-specific analysis. In most cases, the SAM-based interlayer is not a perfectly defined monolayer but a mixed ensemble comprising covalently anchored molecules together with weakly adsorbed species, and the intrinsic stability of this layer becomes a bottleneck for further improving device performance. Importantly, this instability cannot always be addressed through molecular design alone within the conventional SAM layer framework. Going forward, the field may need to move beyond the current design paradigm toward more stable SAM-based molecular architectures that ensure robust anchoring of all molecules. Achieving this would require more in-depth understanding of the complete assembly configuration of SAMs, as well as the dynamic process of molecular assembly.

Although extensive efforts have been devoted to characterizing SAMs, a complete understanding of their stacking microstructure remains elusive, which can partly be attributed to the limited availability of characterization methods that can sensitively probe SAMs, especially at perovskite-buried interfaces, with sufficient chemical specificity and spatial/depth resolution. A practical way forward is to combine complementary techniques that provide spatially and depth-resolved information, enabling joint correlation of anchoring mode, molecular orientation, stacking order, film thickness, and morphology. For instance, surface-sensitive vibrational spectroscopies, such as PM-IRRAS and SFG, can

provide anchoring-group coordination and molecular orientation. These methods could be further extended to spatial mapping to assess lateral heterogeneity, linking microscopic order to macroscopic uniformity. Polarization-modulation infrared reflection-absorption spectroscopy enables quantitative determination of tilt angles and bond orientations with sub-monolayer sensitivity, minimizing substrate interference and making it particularly valuable for analyzing buried interfaces. High-resolution angle-resolved XPS and depth-sensitive profiling, such as ToF-SIMS, could help track molecular migration upon processing or aging stresses. Together, the combination of these methods may facilitate correlation between molecular layer evolution and device degradation behaviors.

Although most current studies interrogate SAMs after the molecular layer has formed, the dynamic self-assembly process remains much less understood. This dynamic process provides a broad, tunable space for controlling molecular assembly. The assembly behaviors are expected to depend on multiple factors, such as substrate chemistry and topography, as well as drying/flow conditions that differ substantially between spin-coating and scalable-coating processes. Even subtle changes in the assembly process may directly translate into distinct final configurations, including coverage continuity, the fraction of covalently anchored species, and the emergence of multilayered or aggregated domains. Future studies may benefit from *in situ* and time-resolved methods, including quartz crystal microbalance with dissipation monitoring (QCM-D) or surface plasmon resonance (SPR) to monitor adsorption in real time and extract characteristic adsorption/reorganization timescales from the time-dependent traces, *in situ* ellipsometry to track thickness evolution, and *in situ* GIWAXS to follow the development of out-of-plane ordering during assembly.

Moreover, most current understanding of SAM layers is derived from spin-coated films in lab-scale. As the field moves toward industrialization with scalable methods such as blade and slot-die coating, it remains unclear whether SAMs follow the same assembly pathways, given the markedly different solvent residence time and drying profile. Future studies may benefit from systematically evaluating how the SAM microstructure, such as molecular ordering, coverage uniformity, and film thickness, evolves across large substrate areas under scalable coating conditions. Establishing such a processing-microstructure map under manufacturing conditions would guide the design of both molecular systems and processing protocols tailored to scalable fabrication.

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DECLARATION OF INTERESTS

The authors declare no competing interests.

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